Report Exercise 1: Launching of Artemis 1

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1 Introduction

Space exploration began with rockets. The first problem they solve is to leave the earth's gravity. With our modern techniques we can model the trajectories from the ground to orbit. This is the ulterior motive that pushes us to start a small example of it.

This study will imagine the rocket Artemis 1 going strait to space. The Euler Explicit method is used to solve the force equation.

2 Calculus and Implementation

The general problem is to simulate the speed of the rocket with non-conservative forces and a full variable mass system. In order to simplify the numerical calculations and to have an intuition, the aerodynamic drag force F_t will also be added or deleted.

2.1 Differential equations for the motion and mass

Newton's second law with variable mass system is expressed as:

$$\begin{aligned} m\ddot{x} &= -F_t - mg + v_{rel}\dot{m} \\ &= -\frac{1}{2}C_x S\dot{x}^2 p_0 e^{-x/\lambda} - mg + v_e D \end{aligned} \tag{1}$$

Where $v_{rel} = v_e$ the ejection velocity of the gas from the rocket, C_x is the drag coefficient, S the cross-sectionals area, x the altitude, λ and p_0 the density coefficients, D the mass flow and g the gravity constant. With m_0 the mass at the beginning, $v = \dot{x}$ and $a = \ddot{x}$ the equation follows:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x \\ v \\ m \end{pmatrix} &= \begin{pmatrix} v \\ a \\ \dot{m} \end{pmatrix} \\ &= \begin{pmatrix} v \\ -\frac{1}{2m}C_x S\dot{x}^2 p_0 e^{-x/\lambda} - g + v_e\frac{D}{m} \\ -D \end{pmatrix} \end{aligned} \tag{2}$$

2.2 Analytic solution

An analytic solution to the equation with $Cx = 0$ may be found by integrating two times the force equation. At $t_0 = 0$ s, the rocket is motionless on the ground ($x_0 = 0$ m, and $\dot{x}_0 = 0$

m/s). By integrating two times from t_0 to t.

$$
\begin{aligned}
\int_{t_0}^{t}\int_{t_0}^{t}\ddot{x} = \int_{t_0}^{t}\int_{t_0}^{t} -g + v_e\frac{D}{m} &= \int_{t_0}^{t}\int_{t_0}^{t} -g + v_e\frac{D}{-Dt+m_0} \\
\int_{t_0}^{t}\dot{x}-\dot{x}_0 = \int_{t_0}^{t} -g(t-t_0) + v_e\log\left(\frac{-Dt+m_0}{-Dt_0+m_0}\right) \\
x = x - x_0 = -\frac{t}{2}(gt-2v_e) - \frac{v_e}{D}(Dt-m)\log\left(1-\frac{Dt}{m_0}\right)
\end{aligned}
\tag{3}
$$

and for the velocity

$$
\begin{aligned}
\int_{t_0}^{t}\ddot{x} = \int_{t_0}^{t} -g + v_e\frac{D}{m} &= \int_{t_0}^{t} -g + v_e\frac{D}{-Dt+m_0} \\
\dot{x}-\dot{x}_0 &= -g(t-t_0) + v_e\log\left(\frac{-Dt+m_0}{-Dt_0+m_0}\right) \\
\dot{x} &= -gt + v_e\log\left(\frac{-Dt}{m_0}+1\right)
\end{aligned}
\tag{4}
$$

2.3 Mechanical energy and power

The gravity may be included in the potential energy. So the mechanical energy may be written:

$$
E_{mec} = E_{cin} = \frac{1}{2}m\dot{x}^2 + mgx \tag{5}
$$

Here is the mechanical energy theorem that has to be verified:

$$
\begin{aligned}
\frac{\mathrm{d}E_{mec}}{\mathrm{d}t} &= (-F_t + v_e D)\cdot\dot{x} \\
&= \frac{\mathrm{d}(\frac{1}{2}m\dot{x}^2 + mgx)}{\mathrm{d}t} \\
&= \frac{1}{2}(\dot{m}\dot{x}^2) + m\dot{x}\ddot{x} + mg\dot{x} + \dot{m}gx \\
\frac{1}{2}(-D\dot{x}^2) + m\dot{x}\ddot{x} + mg\dot{x} - Dgx &= \frac{1}{2}(-D\dot{x}^2) - Dgx - F_t\dot{x} + v_e D\dot{x} \\
&= P_{NC}
\end{aligned}
\tag{6}
$$

So all the forces contribute to the non conservative power:

$$
P_{NC} = \frac{1}{2}(-D\dot{x}^2) - Dgx - F_t\dot{x} + v_e D\dot{x} \tag{7}
$$

3 Simulations and analysis

For all the simulations: $m_0 = 2600000$ kg, $D = 10000$ kg/s, $v_e = 4200$ m/s, $p_0 = 1.2$ kg/m^3, $\lambda = 10000$m, $S = 110$ m^2 and $g = 9.81$ m/s^2. C_x will be modified during the exercise.

3.1 Convergence of the results with $C_x = 0$

With $C_x = 0$ (without air friction), and if the simulation finishes at $t_{fin} = 120$s, then the final velocity and position can be determined analytically. The equation 3 is used for the position and the equation 4 is used for the velocity. One can find: $x_{fin} = 69372.9454571406$ m and $vfin = 1422.76467530614$ m/s.

The number of steps is modified for some simulations ($n_{step} = [160\ 320\ 640\ 1280\ 2560\ 5120\ 10240]$). Then the final position and the final velocity of the rocket is compared to the analytic solution. Now and for the next simulations: $\Delta t(i) = \frac{t_{fin}}{n_{step}(i)}$

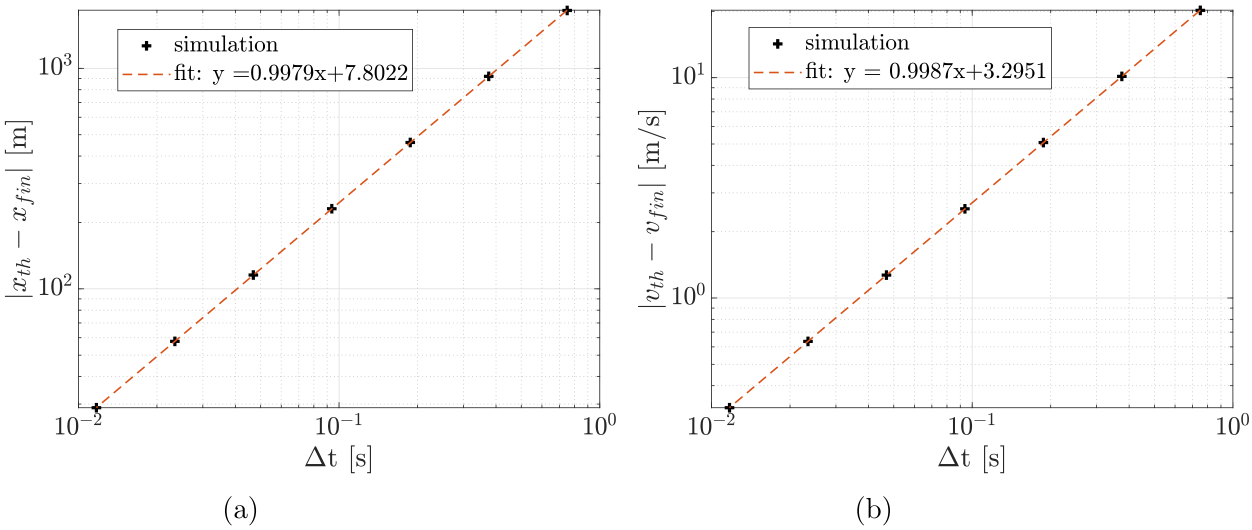


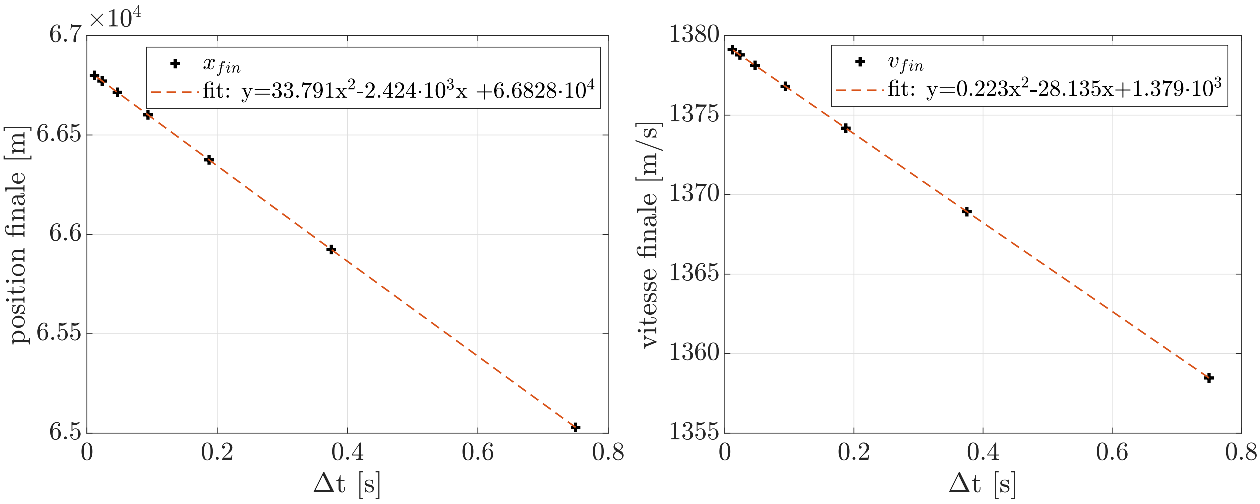
Figure 1: evolution of the error for the final position and the final velocity :
a) position b) velocity

On the two figures the slopes are near 1 so this is a numerical convergence of 1.

3.2 Convergence of the results with $C_x = 0.35$

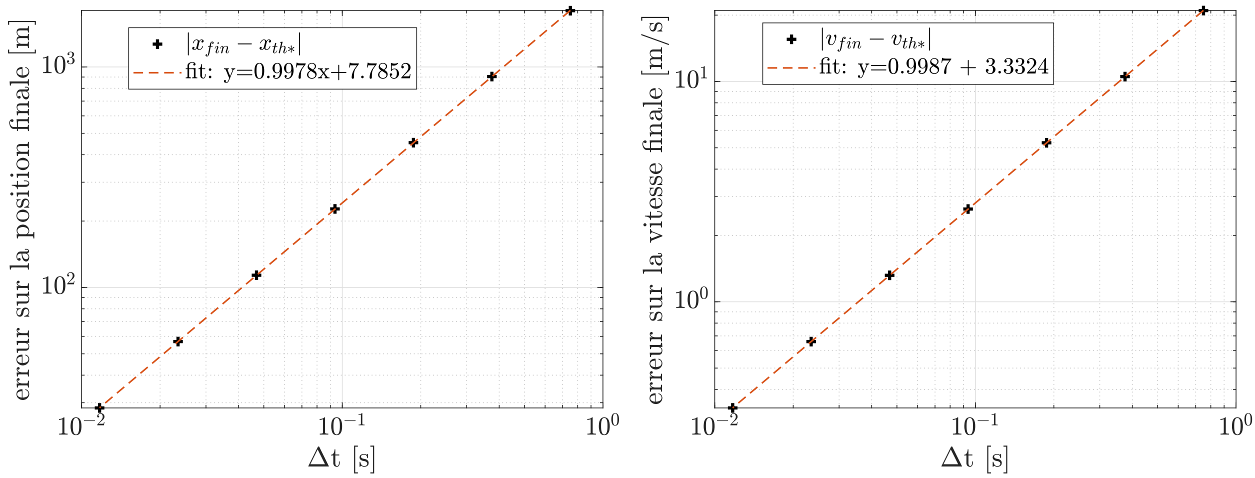
With $C_x = 0.35$ (with air friction), and if the simulation finishes at $t_{fin} = 120$s, then the final velocity and position are determined again. Here we can't have an analytic solution but we can find the final position and velocity when $\Delta t \to 0$. It won't be the exact analytic solution but a good approximation.

The number of steps is modified again: (n_{step} = [160 320 640 1280 2560 5120 10240]).



(a) (b)

Figure 2: evolution of the final position and the final velocity :
a) position b) velocity



(a) (b)

Figure 3: evolution of the error of the final position and the final velocity :
a) position b) velocity

Here, the fit on the velocity and on the position is polynomial and of order 2. To find the final position and velocity: $\Delta t \to 0$. We compute x=0 in the fit and the constant part is the only one that stays: $x_{th*} = 66828$ m and $v_{th*} = 1379$ m/s.

Then as in the first part of the simulations, we compare it to the simulations to build the evolution of the error.

As shown in the fit equation: the order of convergence is 1.

3.3 Evolution of mechanical energy

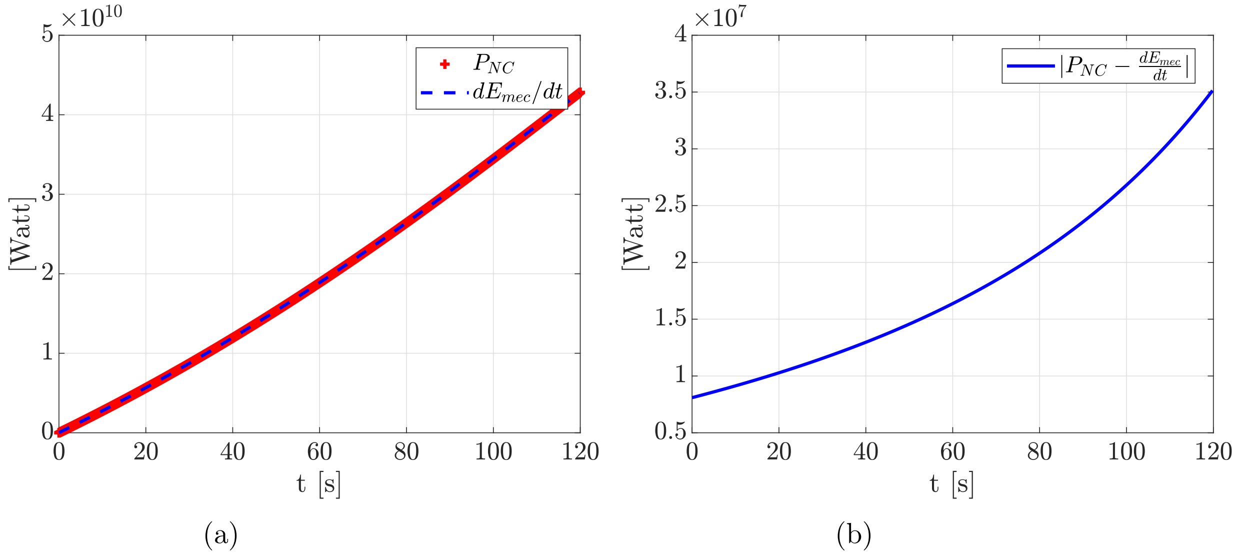


Figure 4: evolution of the the mechanical energy compared to the power of non conservative forces :
a) compared with trajectories b) error between the power and non conservative forces

For this part, the mechanical energy will be compared to the power of non conservative forces with respect to the equations 5 and 7. To verify the mechanical energy theorem (eq. 6), an approximation will be used:

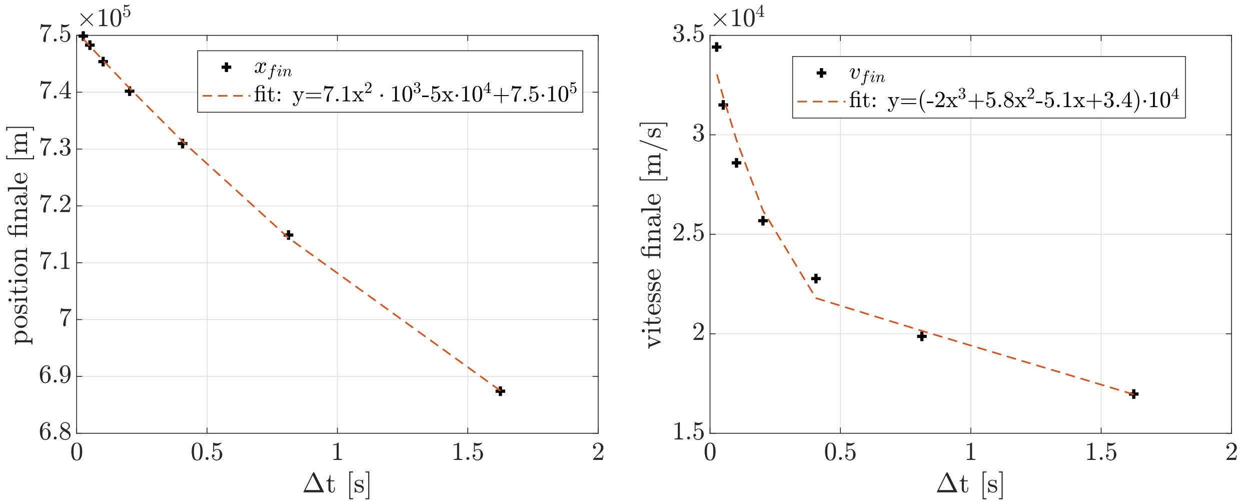
$$\frac{\mathrm{d}E_{mec}(t_j)}{\mathrm{d}t} \approx \frac{E_{mec}(t_{j+1}) - E_{mec}(t_{j-1})}{2\Delta t} \tag{8}$$

Here, the power and $\mathrm{d}E_{mec}(t_j)/\mathrm{d}t$ quite follow each other. The energy is non conservative. This is why we see its derivative non constant.

3.4 Convergence of the results with $t_{fin} = 260[s]$

Here, as in part 2, the order of convergence will be studied with $t_{fin} = 260$ [s]. At this time, all the fuel of the rocket has been burned. So in the equation 3 and 4, the inside of the logarithm part is negative. So the integration is no more possible (or the mass is negative that is not possible). Physically, it might mean that the rocket goes in the other direction. But this is not possible, this is the ground. The rocket should have been given another mass that is without fuel. Here, the mass of the fuel is the mass of the rocket but, in reality, this is not possible. And this is why the graph of the errors on velocity and position is not exactly right.

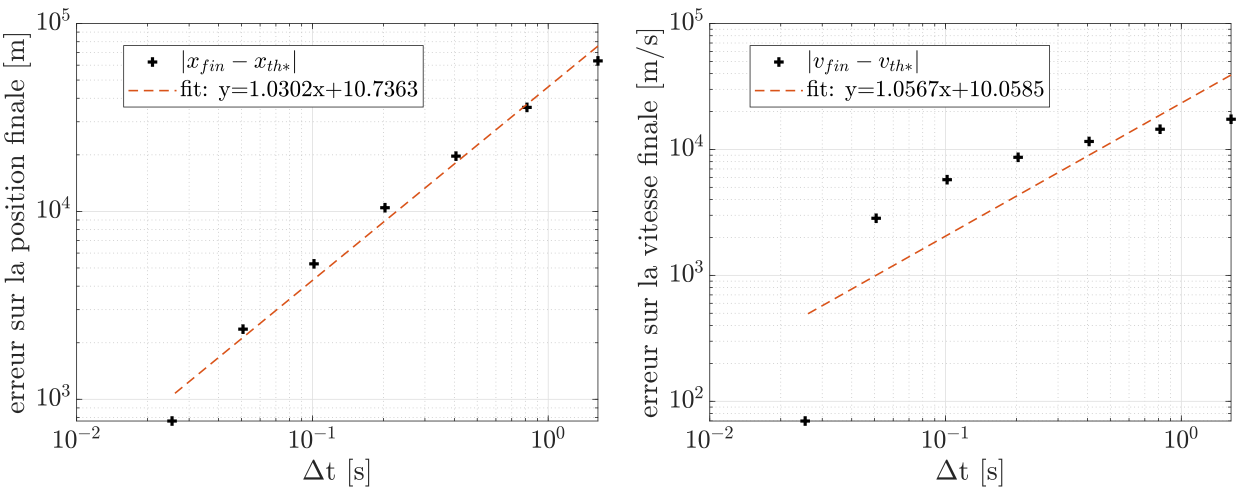
Let's start with $C_x = 0.35$ (with air friction). The order of convergence may be found.



(a) (b)

Figure 5: evolution of the final position and the final velocity :
a) position b) velocity

For x=0, the final position and final mass are: $x_{th*} = 7.5 \cdot 10^5$ m and $v_{th*} = 3.4 \cdot 10^4$ m/s.

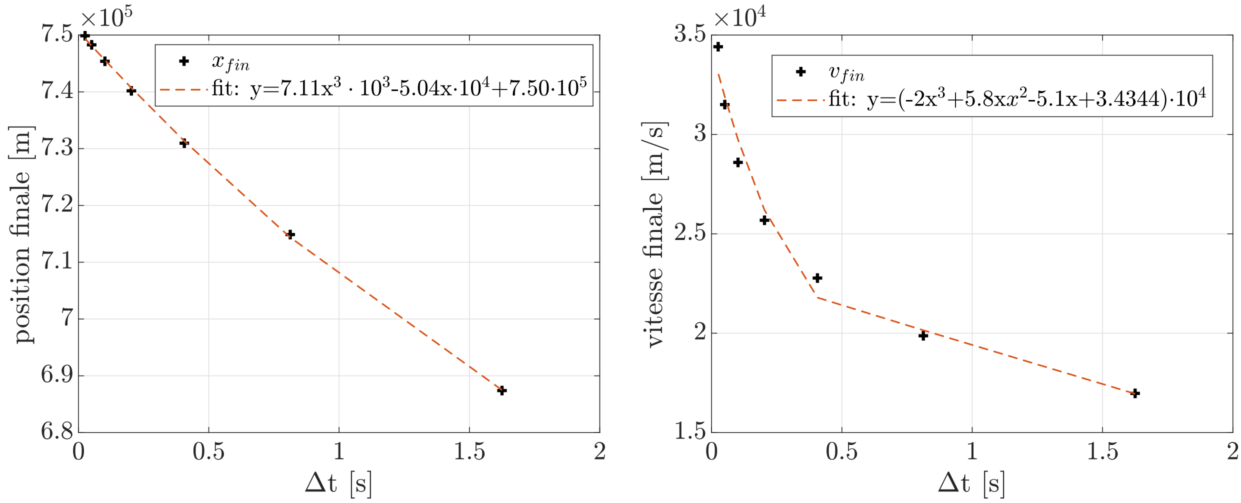


(a) (b)

Figure 6: evolution of the error of the final position and the final velocity :
a) position b) velocity

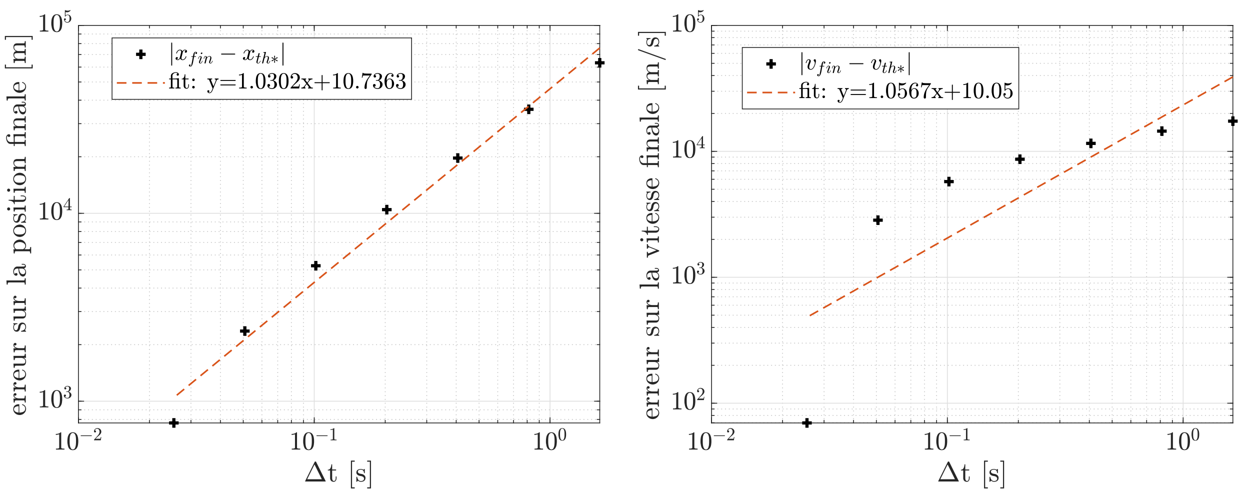
For both velocity and position, the order of convergence is 1.

For $C_x = 0.0$ (without air friction), the analytic solution doesn't exist. The solution is studied the same as above: For x=0, the final position and final mass are: $x_{th*} = 7.5 \cdot 10^5$ m and $v_{th*} = 3.4344 \cdot 10^4$ m/s.



(a) (b)

Figure 7: evolution of the final position and the final velocity :
a) position b) velocity



(a) (b)

Figure 8: evolution of the error of the final position and the final velocity :
a) position b) velocity

The final positions and velocity are the exact same when we have or not friction forces for a big amount of time.

4 Conclusion

In conclusion, in the course of our simulations, we were able to study the velocity and positions of the rocket under various conditions. As discussed in the last part, many improvements may help to make a better approximation of the trajectory of a rocket. But the essential of the mechanisms is here. The understanding of these trajectories is an essential point for the next few years in order to improve our means of communications and to discover the universe.